

TryNex Lifestyle — Complete Product Rebuild Checklist

Status Key: [] Not Started | [~] In Progress | [x] Done | [!] Blocked

PHASE 1 — Complete Audit

- [] Audit frontend storefront (Home, Shop, Product, Cart, Checkout, Design Studio)
- [] Audit backend API server (routes, auth, security, performance)
- [] Audit database schema (indexes, constraints, normalization, relations)
- [] Audit Design Studio / mockup engine (PSD, live preview, composer, 3D)
- [] Audit admin dashboard (analytics, users, orders, products, templates, AI, billing)
- [] Audit mobile app (Expo React Native)
- [] Audit deployment, CI/CD, environment validation, backups
- [] Produce dependency map
- [] Identify every unfinished feature, fake implementation, hardcoded section, broken API, UI inconsistency
- [] Write final audit report with priority matrix

PHASE 2 — Repair Architecture

- [] Refactor duplicated code across frontend and backend
- [] Remove dead code and unused imports/dependencies
- [] Replace fake data with real services
- [] Standardize component architecture (shared UI, hooks, API clients)
- [] Improve folder structure where needed
- [] Create reusable services (storage, image processing, AI, notifications, billing)
- [] Improve state management where needed
- [] Optimize API communication (batching, caching, pagination)

PHASE 3 — PSD Mockup Engine Rebuild

- [] Correct Smart Object detection and replacement
- [] Correct perspective transformation
- [] Correct scaling of artwork to print zones
- [] Correct clipping masks
- [] Correct blend modes (multiply, screen, overlay, etc.)
- [] Correct positioning and rotation
- [] Correct shadows, highlights, and reflections
- [] Correct exports (PNG/JPG/PSD)
- [] Support multiple PSD layers
- [] Support multiple templates per product
- [] No stretched designs
- [] No misaligned artwork
- [] No cropped previews
- [] Generated previews match Photoshop output

PHASE 4 — Live Preview

- [] Real-time updates without refresh
- [] Instant rendering for text, images, logos, patterns, colors
- [] High-resolution preview
- [] No lag during editing
- [] Support all product variants
- [] Support mockup switching

- ☐ Synchronize 2D editor and 3D preview state

PHASE 5 — Product System

- ☐ Real product variants (size, color, material)
- ☐ Real pricing per variant
- ☐ Real inventory tracking
- ☐ Real categories and subcategories
- ☐ Real mockup associations
- ☐ Real previews and thumbnails
- ☐ SEO fields (title, description, keywords, OG, JSON-LD)
- ☐ Product tags
- ☐ Product collections
- ☐ Product search (full-text)
- ☐ Product filters (category, price, color, size, etc.)
- ☐ Product sorting
- ☐ Product pagination

PHASE 6 — AI Integration

- ☐ Fully connect prompt generation
- ☐ Fully connect image generation (all models)
- ☐ Fully connect background removal (server + browser fallback)
- ☐ Fully connect upscaling
- ☐ Fully connect mockup creation from AI images
- ☐ Product title generation from prompts/images
- ☐ SEO description generation
- ☐ Keyword and tag generation
- ☐ Alt text generation
- ☐ Metadata generation for generated assets
- ☐ AI usage tracking and credits

PHASE 7 — Admin Dashboard

- ☐ Complete analytics dashboard (sales, orders, users, products)
- ☐ Users management (list, search, roles, ban/unban)
- ☐ Orders management (list, filter, status update, messages)
- ☐ Products management (CRUD, variants, pricing, inventory, SEO)
- ☐ Templates / PSD library management
- ☐ Storage management (view usage, delete, move)
- ☐ Logs viewer (activity, errors, AI usage)
- ☐ AI usage and credits tracking
- ☐ Billing and invoices
- ☐ Error monitoring
- ☐ System monitoring and health

PHASE 8 — Database

- ☐ Normalize schema
- ☐ Add missing indexes on foreign keys and filter columns
- ☐ Add missing constraints (foreign keys, check constraints, ranges)
- ☐ Convert status/role text columns to enums or check constraints
- ☐ Write proper migrations (up and down)
- ☐ Define relations and foreign keys explicitly
- ☐ Use transactions for multi-step operations
- ☐ Set up automated backups and backup verification

- ☐ Document schema and relationships

PHASE 9 — Security

- ☐ Input validation on every API endpoint (Zod schemas)
- ☐ Rate limiting on all public and admin endpoints
- ☐ CSRF protection for state-changing requests
- ☐ XSS prevention (output encoding, CSP headers)
- ☐ SQL injection prevention (parameterized queries / Drizzle)
- ☐ Strong authentication (Clerk or Replit Auth preferred)
- ☐ Authorization checks (RBAC) on admin routes
- ☐ Secure environment variables and secrets management
- ☐ Encryption at rest and in transit
- ☐ Security audit and penetration test pass

PHASE 10 — Performance

- ☐ Lazy load routes and heavy components
- ☐ Implement caching strategy (Redis, browser, CDN)
- ☐ Image optimization (WebP/AVIF, responsive sizes, lazy loading)
- ☐ Code splitting and tree shaking
- ☐ Compression (gzip/brotli)
- ☐ Bundle optimization
- ☐ Query optimization (DB indexes, pagination, limits)
- ☐ Memory optimization (no leaks in Design Studio)
- ☐ Background jobs for heavy tasks (image processing, exports, AI)
- ☐ Performance budgets and monitoring

PHASE 11 — Responsive Design

- ☐ Perfect desktop layout
- ☐ Perfect laptop layout
- ☐ Perfect tablet layout
- ☐ Perfect mobile layout
- ☐ Every page responsive
- ☐ No horizontal overflow
- ☐ No broken layouts on any breakpoint
- ☐ Touch-friendly Design Studio on mobile
- ☐ Accessible navigation and focus states

PHASE 12 — UI/UX

- ☐ Premium SaaS appearance
- ☐ Consistent spacing and typography
- ☐ Smooth animations and transitions
- ☐ Loading states and skeleton loaders
- ☐ Toast notifications for all user actions
- ☐ Modern card and dashboard design
- ☐ Professional forms with validation
- ☐ Accessible colors (WCAG contrast)
- ☐ Empty states and error states
- ☐ Onboarding and help tooltips

PHASE 13 — Testing

- ☐ Test every API endpoint (automated)

- ☐ Test every page (render, navigation, error boundaries)
- ☐ Test every button and form
- ☐ Test every modal and dialog
- ☐ Test every upload and download flow
- ☐ Test every AI feature
- ☐ Test every PSD template and mockup
- ☐ Test every database operation
- ☐ Test every authentication and authorization flow
- ☐ Fix every failure automatically
- ☐ Repeat until zero critical issues remain

PHASE 14 — Deployment

- ☐ Production-ready build pipeline
- ☐ Environment validation on startup
- ☐ Health checks (DB, Redis, storage, AI services)
- ☐ Structured logging and log aggregation
- ☐ Monitoring (uptime, performance, errors)
- ☐ Error reporting integration
- ☐ Automatic backups (DB + assets)
- ☐ Optimized production build (minified, compressed, no source maps)
- ☐ Staging environment parity
- ☐ Rollback plan documented and tested
- ☐ No placeholders anywhere
- ☐ No TODO comments left in code
- ☐ No unfinished code paths
- ☐ No simulated functionality
- ☐ No mock APIs in production
- ☐ No fake previews
- ☐ Everything works using real code
- ☐ Full `pnpm run typecheck` passes
- ☐ Full build passes for every artifact
- ☐ All critical tests pass
- ☐ Project documented in `PROJECT\_REBUILD\_TRACKER/`

Last Updated: 2026-07-22 | Next: Complete Phase 1 audit